

Modelling the Internal Structure of a Zero-Age Main Sequence Star

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ABSTRACT

In this work, I present a Python-based code designed to model the interior structure of a zero-age main sequence (ZAMS) star in the form of luminosity, pressure, radius, and temperature all as a function of mass. Based on these calculations, it was determined that a $1.2M_{\odot}$ star would have a luminosity of 8.12×10^{33} erg/s, a central pressure of 1.38×10^{17} dyne/cm², a radius of 1.19×10^{11} cm, and an effective temperature of 5,324 K. To verify the accuracy of the model, I compare the results to the model output of a star of the same specifications by the Modules for Experiments in Stellar Astrophysics (MESA) program. All physical quantities had a percent difference of 15 – 40% indicating some inaccuracies that could be due to assumptions made in the calculation process. The code is publicly available on GitHub^{a)}.

1. INTRODUCTION

Stars are historically one of the most studied objects in astronomy. Being one of the most easily visible objects, past civilizations could use them as tools to track the seasons or navigate ships. They have always been observed externally, but only recently have we been able to probe their internal structures. Computers and numerical integration have allowed for the simulation of their internal structures, which cannot be observed directly. We have been able to achieve highly accurate stellar models using computational methods, such as with Modules for Experiments in Stellar Astrophysics (MESA) (Paxton et al. (2010)). In this work, I present a model, achieved using more basic assumptions than MESA, of the internal structure of a $1.2 M_{\odot}$, solar metallicity, zero-age main sequence (ZAMS) star. In section 2, I discuss the physical problem and the equations and assumptions used for the calculation. In section 3, I introduce the model used and calculations performed. In section 4, I analyze the results of the model. And in section 5, I compare the results obtained from my model with that obtained from MESA.

2. PHYSICAL PROBLEM: PARAMETERS AND ASSUMPTIONS

There are four coupled differential equations that can be used to describe the interior of a star (Hansen et al.

(2004), Kippenhahn et al. (2012)). They can be written in terms of the Lagrangian mass coordinate M_r and describe hydrostatic equilibrium (1), mass conservation (2), energy generation (3), and radiative transport (4) and operate under the assumption of spherical symmetry:

$$\frac{dP}{dM_r} = -\frac{GM_r}{4\pi r^4} \quad (1)$$

$$\frac{dr}{dM_r} = \frac{1}{4\pi r^2 \rho} \quad (2)$$

$$\frac{dL_r}{dM_r} = \epsilon = \epsilon_{pp} + \epsilon_{CNO} \quad (3)$$

$$\frac{dT}{dM_r} = \frac{GM_r T}{4\pi r^4 P} \nabla \quad (4)$$

where P is the pressure (dyne cm⁻²), r is the radius (cm), L is the luminosity (erg s⁻¹), T is the temperature (K), ϵ is the energy generation rate per unit mass (erg g⁻¹ s⁻¹), and ∇ is the Logarithmic gradient $\frac{d \ln T}{d \ln P}$.

2.1. Equation of State

Since we are modelling a ZAMS, we can assume homogeneous composition of an ideal, fully ionized gas. Because of these assumptions, the mean molecular weight μ will be constant with a value determined by

$$\mu \approx \frac{4}{6X + Y + 2} \quad (5)$$

where X is the hydrogen mass fraction. For a typical ZAMS, $\mu \approx 0.6$ (Hansen et al. (2004)). Under the assumptions of local thermodynamic equilibrium (LTE),

^{a)} <https://github.com/evelynbruinsma/StellarStructure>

the equation of state can be a combination of ideal gas and radiation, as described in [Hansen et al. \(2004\)](#):

$$P = \frac{\rho N_A k T}{\mu} + \frac{a T^4}{3} = P_g + P_{rad} \text{ dyne/cm}^2 \quad (6)$$

where N_A is Avogadro's number, k is the Boltzmann constant, a is the radiation constant $a = \frac{4\sigma}{c}$, σ is the Stefan-Boltzmann constant, and c is the speed of light.

2.2. Opacity

The opacity κ measures a material's ability to block photons and can be affected by processes such as electron scattering, free-free absorption, bound-free absorption, bound-bound absorption, H^- opacity, and other sources. Due to the complexity of calculating true stellar opacities, this model utilizes existing opacity tables from the Opacity Project². These tables can be interpolated for specific values of density and temperature to determine the opacity.

2.3. Energy Generation

A star's energy generation comes from the burning of hydrogen into helium and takes place in its core. This can occur via the proton-proton chain (pp-chain) or the carbon-nitrogen-oxygen (CNO) cycle. The pp-chain, which converts protons into helium nuclei, is dominant in smaller stars, like the Sun, as it operates at lower temperatures. On the other hand, the CNO cycle, which involves using carbon, nitrogen, and oxygen as catalysts to form helium, dominates in more massive stars due to its higher temperature requirements. The total energy generation rate is determined by the sum of the constituents from both the pp-chain and the CNO cycle, both found in chapter 18 of [Kippenhahn et al. \(2012\)](#):

$$\epsilon_{tot} = \epsilon_{pp} + \epsilon_{CNO} \quad (7)$$

In the case of the pp-chain,

$$\epsilon_{pp} = 2.57 \times 10^4 \psi f_{11} g_{11} \rho X_1^2 T_9^{-2/3} e^{(-3.381/T_9^{1/3})} \quad (8)$$

$$g_{11} = (1 + 3.82T_9 + 1.51T_9^2 + 0.144T_9^3 - 0.0114T_9^4) \quad (9)$$

where f_{11} is the shielding factor and ψ is a correction factor to account for additional energy generation in the pp2 and pp3 branches. In the case of the CNO cycle,

$$\epsilon_{CNO} = 8.24 \times 10^{25} g_{14,1} X_{CNO} X_1 \rho T_9^{-2/3} e^{(-15.231T_9^{-1/3} - (T_9/0.8)^2)} \quad (10)$$

$$g_{14,1} = (1 - 2.00T_9 + 3.41T_9^2 - 2.43T_9^3) \quad (11)$$

² OP Database: <https://cds.unistra.fr/topbase/testop/home.html>

2.4. Energy Transport

Within a ZAMS, energy can be transported radiatively or convectively depending on the value of ∇ . The radiative gradient is:

$$\nabla_{rad} \equiv \left(\frac{d \ln T}{d \ln P} \right)_{rad} \equiv \frac{3}{16\pi a c} \frac{P \kappa}{T^4} \frac{L}{GM} \quad (12)$$

where κ is the Rosseland mean opacity. The adiabatic gradient is given by:

$$\nabla_{ad} = \frac{\Gamma_2 - 1}{\Gamma_2} \quad (13)$$

For an ionized gas, $\Gamma = \frac{5}{3}$, which yields $\nabla_{ad} = 0.4$. Both ∇_{rad} and ∇_{ad} are compared, and the method of energy transport is determined by the smaller value. For radiative energy transfer, $\nabla = \nabla_{rad} < \nabla_{ad}$. For convection to occur, $\nabla = \nabla_{ad} < \nabla_{rad}$.

3. NUMERICAL CALCULATION

The system of coupled differential equations listed in 2 must be solved using numerical integration, which can be done using the model parameters and assumptions listed above.

3.1. Shooting Function

The differential equations are integrated numerically using the Runge-Kutta RK45 method within scipy's `solve_ivp` function. Due to the different boundary conditions, discussed in 3.2, the equations must be solved twice, once from the surface inward and once from the center outward, where they meet at an intermediary mass point. This can be achieved using a shooting function such as that discussed in [Press et al. \(1988\)](#). This type of method finds a set of solutions that satisfy the boundary conditions both at the center and the surface of the star. The best fit input parameters for these solutions were calculated with `scipy.optimize.minimize`.

3.2. Boundary Conditions

3.2.1. Center Boundary Conditions

To calculate the solutions for the center integration of the star, I start at a mass point ever so slightly away from the center (corresponding to $M_{r,c} = 10^{-5} M_\odot$), which then determines the central luminosity ($L_c = \epsilon M_{r,c}$). A central density ρ_c is calculated based on the central pressure P_c and central temperature T_c , both provided as initial guesses. As described in [Kippenhahn et al. \(2012\)](#), the boundary conditions become:

$$L_{c'} = M\epsilon(T_c, \rho_c) \quad (14)$$

$$R_{c'} = \left(\frac{3M_r}{4\pi\rho_c} \right)^{1/3} \quad (15)$$

$$P_{c'} = P_c - \frac{3G}{8\pi} \left(\frac{4\pi\rho_c}{3} \right)^{4/3} M_r^{2/3} \quad (16)$$

$$T_{c'}^4 = T_c^4 - \frac{1}{2ac} \left(\frac{3}{4\pi} \right)^{2/3} \kappa_c \epsilon_c \rho_c^{4/3} M_r^{2/3} \quad (17)$$

$$\ln T_{c'} = \ln T_c - \frac{\pi^{1/3} G \nabla_{ad,c} \rho_c^{4/3} M_r^{2/3}}{6 P_c} \quad (18)$$

17 corresponds to a radiative core, while 18 corresponds to a convective core. Since $\sim 1.2M_\odot$ is the boundary between convective and radiative cores (Maeder (2008)), the model is equipped to handle either situation.

3.2.2. Surface Boundary Conditions

Modelling the true surface conditions of a star is difficult and beyond the scope of this course, so I have taken a simplified approach. We cannot assume the surface is where the pressure and temperature go to zero because, in reality, there is a gradient as we exit the star and approach the surrounding material. However, we can consider the pressure at the photosphere ($\tau = 2/3$). As discussed in chapter 4.3 of Hansen et al. (2004), the pressure becomes:

$$P(\tau_p) = \frac{2}{3} \frac{g_s}{\kappa_p} \left(1 + \frac{\kappa_p L}{4\pi c G M} \right) \quad (19)$$

where $g_s = \frac{GM}{R^2}$ is the surface gravity, and the second factor in the parenthesis can be ignored except in the case of massive and luminous stars. This leaves us to calculate the following surface boundary conditions:

$$L = L \quad (20)$$

$$P = \frac{2}{3} \frac{GM}{R^2} \frac{1}{\kappa_p} \quad (21)$$

$$R = R \quad (22)$$

$$T = T_{eff} = \left(\frac{L}{4\pi R^2 \sigma_{sb}} \right)^{1/4} \quad (23)$$

where R and L are passed to the function. For simplicity, a standard κ_p value of 0.34 was chosen (see 4.2 for discussion).

3.3. Initial Guesses

My initial guesses for luminosity and radius were based on the homology relations from section 1.6 of Hansen et al. (2004):

$$\frac{R}{R_\odot} \approx \left(\frac{M}{M_\odot} \right)^{0.75} \quad (24)$$

$$\frac{L}{L_\odot} \approx \left(\frac{M}{M_\odot} \right)^{3.5} \quad (25)$$

The initial guesses for central pressure and temperature were based on the constant density model from section 1.4 of Hansen et al. (2004):

$$P_c = \frac{3}{8\pi} \frac{GM^2}{R^4} \quad (26)$$

$$T_c = \frac{1}{2} \frac{GM}{R} \frac{\mu}{N_A k} \quad (27)$$

However, the constant density model is generally considered to be a poor model for a star, and after further inspection, these parameters yielded very poor results. I then switched the central pressure and temperature to be the solar values taken from the NASA Sun Fact Sheet³. All four initial guesses were then multiplied by a scale factor, found through trial and error, to get both the initial and final results closer to convergence, with the individual scale factors falling between 1.25-1.5.

4. RESULTS

4.1. Initial Guesses

The model was first run with the initial guesses discussed in 3.3, with the full values listed in Table 1.

Table 1. Initial Parameter Guesses

Luminosity (erg/s)	9.05×10^{33}
Central Pressure (dyne/cm ²)	2.73×10^{17}
Radius (cm)	1.02×10^{11}
Central Temperature (K)	2.23×10^7

The results of the stellar structure with these parameters are shown in Figure 1. The inner and outer integrations do not meet at the mid point and require a best fit to be found.

4.2. Best Fit

Since the initial guesses yielded a poorly fit model, I had to find the best fit parameters using the shooting method described in 3.1 as well as

³ <https://nssdc.gsfc.nasa.gov/planetary/factsheet/sunfact.html>

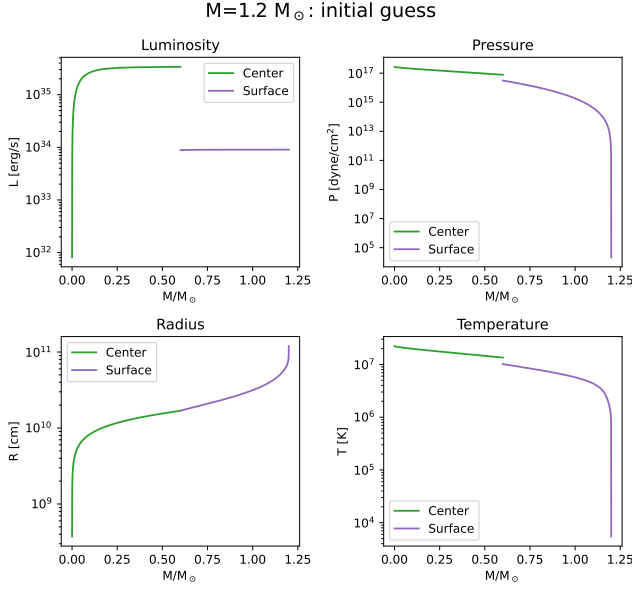


Figure 1. Model Generated from Initial Guesses

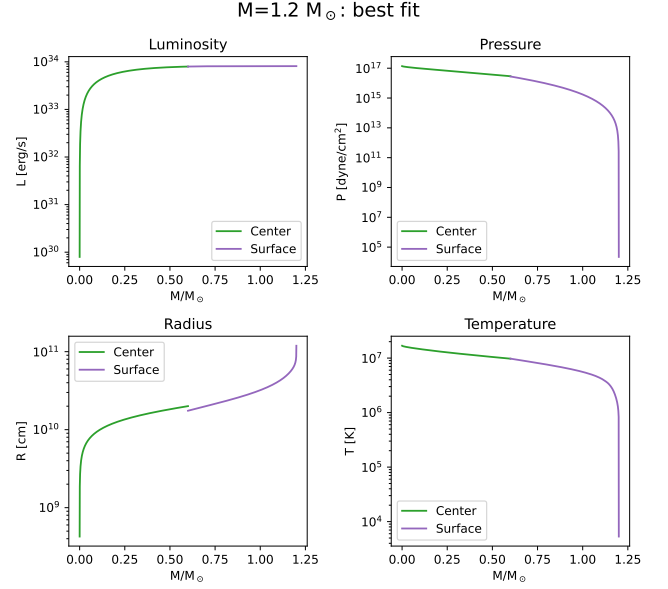


Figure 2. Model Generated from Best Fit Parameters

Table 2. Best Fit Parameters

Luminosity (erg/s)	8.12×10^{33}
Central Pressure (dyne/cm ²)	1.38×10^{17}
Radius (cm)	1.19×10^{11}
Central Temperature (K)	1.68×10^7

scipy.optimize.minimize. After 56 iterations, I received the best fit parameters, listed in Table 2.

I was able to get the model to run in under 50 iterations by using a smaller initial mass value in the center loading function, but the resulting model was even further from convergence. I decided to have it run more iterations in order to present a more accurate model.

The model using the best fit parameters is shown in Figure 2. Figure 3 shows the normalized model. The luminosity converges completely, and the pressure and temperature come very close. The two solutions do touch in the middle, but there is not a smooth transition between the two. The radius, however, does not converge at all.

The lack of convergence could be related to some of the earlier assumptions I made, one of which being the assumption of a fixed κ value in the surface boundary conditions (see 3.2.2) rather than taking into account the Eddington grey opacity mentioned in section 4.3 of Hansen et al. (2004). Another potential source of error could be related to assumptions made when calculating the energy generation rate for the proton-proton chain. Kippenhahn et al. (2012) describes a correction factor ψ to correct for any additional energy produced from

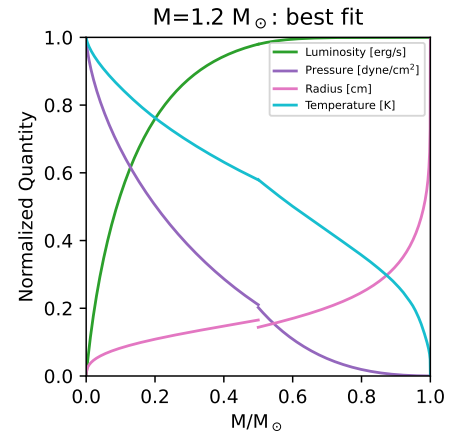


Figure 3. Normalized Best Fit Model

the pp2 and pp3 branches. I assumed a value of $\psi = 1$ for simplicity, which could affect the model. I also used the same amount of steps for both integrations, and it is possible they needed to be treated separately.

Based on the output and values of ∇ , the star has a convective core and a radiative envelope, which aligns with Maeder (2008), which states that the transition between radiative and convective cores is around $1.2M_{\odot}$.

5. COMPARISON WITH MESA

I recreated the same $1.2M_{\odot}$, solar-composition star using MESA to determine the accuracy of my model. Figure 4 shows the MESA results plotted on top of the best fit model. The two models follow each other closely until they hit the surface, where they diverge slightly.

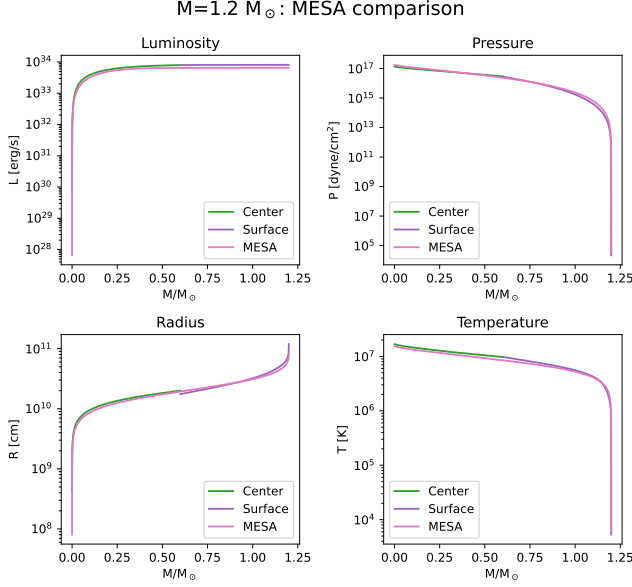


Figure 4. Comparison of Model with MESA

Table 3. Comparison with MESA Parameters

Parameter	Model	MESA	Percent Difference
Luminosity (erg/s)	8.12×10^{33}	6.53×10^{33}	21.66%
Central Pressure (dyne/cm 2)	1.38×10^{17}	1.77×10^{17}	25.03%
Radius (cm)	1.19×10^{11}	7.99×10^{10}	39.38%
Effective Temperature (K)	5324	6176	14.82%
Surface Gravity (cm/s 2)	1.12×10^4	2.49×10^4	75.82%

Again, this could be due to my assumption of a fixed opacity at the surface rather than considering the Eddington grey opacity.

The percent difference between the two models (in surface luminosity, central pressure, radius, central temperature, and surface gravity) is shown in Table 3 and was calculated using:

$$\text{Percent Difference} = \frac{\|X_2 - X_1\|}{\frac{X_2 + X_1}{2}} \times 100 \quad (28)$$

Based on the percentage difference, my model came closest to predicting the effective temperature of the star, and it struggled the most with predicting the radius and therefore the surface gravity, since $g \propto R^{-2}$.

6. CONCLUSION

In this work, I modelled the interior structure of a $1.2M_{\odot}$, solar composition star represented by four couple differential equations under the assumptions of being in hydrostatic equilibrium, having spherical geometry, being composed of a fully-ionized gas, and being in LTE. The energy is transported via radiation and convection, and the method of energy generation comes from the pp-chain and the CNO cycle. The opacity is generated from an interpolation of data from the Opacity Project. The differential equations are solved as a boundary conditions problem with the boundaries residing at the surface and center of the star. They were integrated using a Runge-Kutta method via a shooting method, and the solutions are then optimized to determine best fit parameters. The resulting stellar quantities based on the best fit parameters are then compared to the results from MESA, all of which is listed in Table 3. Based on the results of this comparison, my model is not very effective at modelling the interior of a star and should only be used as a rough estimate.

7. SOURCE CODE

The code used to calculate this model is publicly available at: <https://github.com/evelynbruinsma/StellarStructure>.

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